

Business Intelligence Insights & Storytelling Report

1. Introduction

In the context of the rapid growth of digital payments, fintech companies are required to process large volumes of transactional data while ensuring security, efficiency, and profitability. This Business Intelligence project focuses on a PayPal-like fintech use case, aiming to analyze transaction data in order to extract actionable insights that support business decision-making.

The project was conducted as an end-to-end BI solution, covering the entire analytical lifecycle: business understanding, data preparation, data modeling, visualization, and insight generation. The dataset used contains more than 10,000 transaction records, making it suitable for meaningful analysis and performance evaluation.

The main objective of this project is to:

- Understand transaction behavior across time, geography, customers, and merchants
- Identify patterns related to transaction volume, revenue, and fraud
- Provide data-driven recommendations to improve operational efficiency and risk management

2. Data Preparation and Modeling Overview

The raw transaction dataset underwent a comprehensive ETL (Extract, Transform, Load) process. Data cleaning included handling missing values, removing duplicates, correcting data types, standardizing categorical values, and filtering invalid transaction amounts. A clean dataset was produced and documented through a detailed data dictionary to ensure clarity and transparency.

For analytical efficiency, the data was structured using a dimensional star schema. A central Fact_Transactions table stores quantitative metrics such as transaction amount, fees, and fraud indicators. This fact table is connected to multiple dimension tables, including:

- Dim_Date for time-based analysis
- Dim_Customer for customer segmentation
- Dim_Merchant for merchant behavior analysis
- Dim_Country for geographic insights

This model enables flexible and performant analysis across multiple business perspectives and supports dashboard development.

3. Key Findings

Based on the dashboards and analytical exploration, several important insights were identified:

Finding 1: Transaction activity is highly time-dependent

Transaction volumes peak during standard business hours, particularly between late morning and early afternoon. This suggests that most users perform transactions during working hours, likely linked to online purchases and business-related payments.

Finding 2: High-value transactions are more prone to fraud

Fraudulent transactions are more frequent among higher transaction amounts. While low-value transactions occur more often, fraud risk increases significantly as transaction value increases, highlighting the importance of targeted fraud monitoring.

Finding 3: A small number of countries dominate transaction volume

The analysis shows that a limited number of countries generate the majority of transactions and revenue. This concentration indicates strong market presence in specific regions and potential dependency on key geographic markets.

Finding 4: Card payments are the most used payment method

Card-based transactions represent the largest share of payments. However, they also exhibit a higher fraud rate compared to other payment methods, making them a critical focus area for risk management.

Finding 5: Transaction fees contribute significantly to revenue

Transaction fees represent an important source of revenue, especially during high-volume periods. Small fee adjustments can therefore have a substantial impact on overall profitability.

Finding 6: Fraud cases are unevenly distributed across merchants

Certain merchants show a higher frequency of fraudulent transactions, suggesting the need for merchant-level risk profiling and differentiated monitoring strategies.

4. Why These Results Matter

These findings provide valuable insights into both operational performance and risk exposure. Understanding peak transaction periods allows better system capacity planning and customer support allocation. Identifying high-risk transactions and payment methods supports more efficient fraud detection strategies, reducing financial losses and improving customer trust.

Geographic concentration insights help the company prioritize strategic markets while also identifying opportunities for expansion into underrepresented regions. Moreover, understanding the relationship between transaction volume and fee revenue supports pricing optimization decisions.

Overall, these results enable the organization to move from reactive decision-making to a data-driven, proactive approach.

5. Business Recommendations

Based on the insights derived from the analysis, the following business recommendations are proposed:

1. **Strengthen fraud monitoring for high-value transactions**
Apply stricter validation rules and real-time checks for transactions above certain monetary thresholds.
2. **Implement enhanced security measures for card payments**
Introduce additional authentication steps, such as two-factor authentication, for card-based transactions.
3. **Develop merchant risk profiles**
Classify merchants based on historical fraud rates and apply differentiated monitoring policies.
4. **Optimize system performance during peak hours**
Allocate technical and customer support resources more efficiently during high-activity periods.
5. **Focus marketing and partnerships on high-performing countries**
Reinforce presence in top-performing regions while exploring growth opportunities in emerging markets.
6. **Review transaction fee strategies**
Analyze fee elasticity and consider dynamic pricing models to maximize revenue without harming user satisfaction.
7. **Invest in advanced fraud analytics**
Incorporate machine learning models to predict fraud patterns and enhance detection accuracy.

6. Limitations and Future Improvements

Despite the value of the analysis, several limitations should be acknowledged. The dataset is limited to transactional information and does not include detailed

customer demographics or behavioral data. Additionally, the data represents historical transactions only and does not capture real-time dynamics.

Future improvements could include:

- Integrating customer demographic and behavioral data
- Incorporating real-time streaming data for live fraud detection
- Applying predictive analytics and machine learning techniques
- Expanding the model to include merchant performance and customer lifetime value

These enhancements would further strengthen the BI solution and increase its strategic impact.

7. Conclusion

This project successfully demonstrates the implementation of a complete Business Intelligence solution within a fintech context. Through structured data preparation, dimensional modeling, dashboard development, and insightful analysis, the project delivers meaningful business value.

The insights generated highlight key transaction patterns, fraud risks, and revenue drivers, enabling informed decision-making. By translating data into actionable recommendations, this BI solution supports operational efficiency, risk mitigation, and long-term strategic planning.